10. Decoupling

Decoupled systems

so my plan for the lecture is, first,

to work out, in some detail, a specific example where

decoupling is done to show how that leads to the solution.

And then we'll go back and see how

to do it in general because you will

see that, as I do the decoupling in this particular example,

that that particular method, though,

it's suggested it will not work in general.

Let us go back to homogeneous systems to discuss the concept of “decoupling".

A homogeneous system of ODEs is called **decoupled** if each equation involves only one variable (and if all coefficients are real). In matrix form, this means a homogeneous system is **decoupled** if the coefficient matrix $\mathbf{D}$ is **diagonal** and **real**:

$$\dot{\mathbf{u}} = \mathbf{D}\mathbf{u} \tag{4.164}$$

A decoupled system can be solved by solving each differential equation separately. For example, in the 3×3 case

$$\dot{\mathbf{u}} = \begin{pmatrix} k_1 & 0 & 0 \\ 0 & k_2 & 0 \\ 0 & 0 & k_3 \end{pmatrix} \mathbf{u}$$

$$\Longleftrightarrow$$

$$\begin{aligned} \dot{u}_1 &= k_1 u_1 \\ \dot{u}_2 &= k_2 u_2 \\ \dot{u}_3 &= k_3 u_3 \end{aligned}$$

The general systems

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$$

we have been dealing with are **coupled,** since the matrix $\mathbf{A}$ is not in general diagonal, and at least some of the differential equations involve more than one variable.

Decoupling means performing a linear change of variable so that in the new variables, the system is decoupled and hence each equation can be solved separately. In other words, find $\mathbf{u} = \mathbf{B}\mathbf{x}$ so that $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ becomes $\dot{\mathbf{u}} = \mathbf{D}\mathbf{u}$ when written in terms of the new variables.

How do we find the decoupling variables? One way is to use physical considerations, as shown in the video below.

Another way, the general method, involves diagonalizing the matrix $\mathbf{A}$. We will discuss the general method on the next page.

Decoupling by physical considerations

So that's the solution written out in terms of x and y,

either as a vector in the usual way or in terms

separately in terms of x and y.

But notice in order to do that, you

have to have the equations, these backwards equations.

In other words, I need the equations.

In that form, I need the equations

because they tell me what the new variables are.

But I also have to have the equations the other way

in order to get the solution in terms of x and y finally.

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